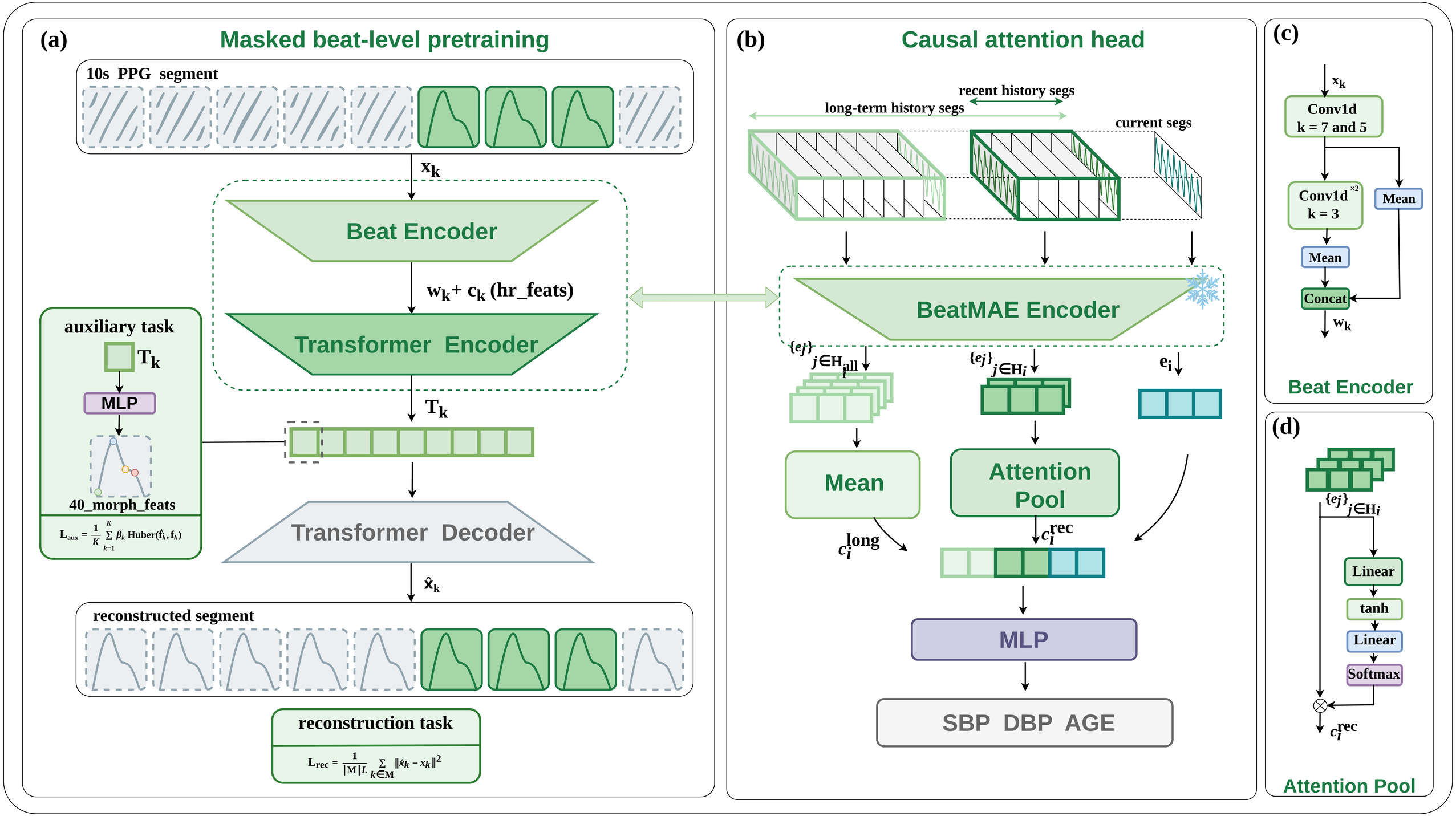

(a)
Masked beat-level pretraining
10s PPG segment
x_k
Beat Encoder
$w_k + c_k$ (hr_feats)
Transformer Encoder
T_k
auxiliary task
T_k
MLP
40_morph_feats
$L_{aux} = \frac{1}{K} \sum_{k=1}^{K} \beta_k \text{Huber}(\hat{f}_k, f_k)$
Transformer Decoder
$\hat{x}_k$
reconstructed segment
reconstruction task
$L_{rec} = \frac{1}{|M|L} \sum_{k \in M} \|\hat{x}_k - x_k\|^2$
(b)
Causal attention head
recent history segs
long-term history segs
current segs
BeatMAE Encoder
$\{e_j\}_{j \in H_{all}}$
$\{e_j\}_{j \in H_i}$
e_i
Mean
Attention Pool
c_i^{long}
c_i^{rec}
MLP
SBP DBP AGE
(c)
x_k
Conv1d k = 7 and 5
Conv1d k = 3
×2
Mean
Mean
Concat
w_k
Beat Encoder
(d)
$\{e_j\}_{j \in H_i}$
Linear
tanh
Linear
Softmax
c_i^{rec}
Attention Pool